

2.7 Carrier Generation & Recombination

- Perturbation of semiconductor, i.e. an **excess or deficit in carrier concentration** with respect to the equilibrium values is created, resulting in **non-equilibrium conditions**
- If the perturbation is removed, recombination/generation (R-G) processes will restore equilibrium conditions; if perturbation is maintained, R-G processes will stabilize the (non-equilibrium) carrier concentrations

2.7.1 Generation and Recombination Processes

2.7.2 R-G Statistics

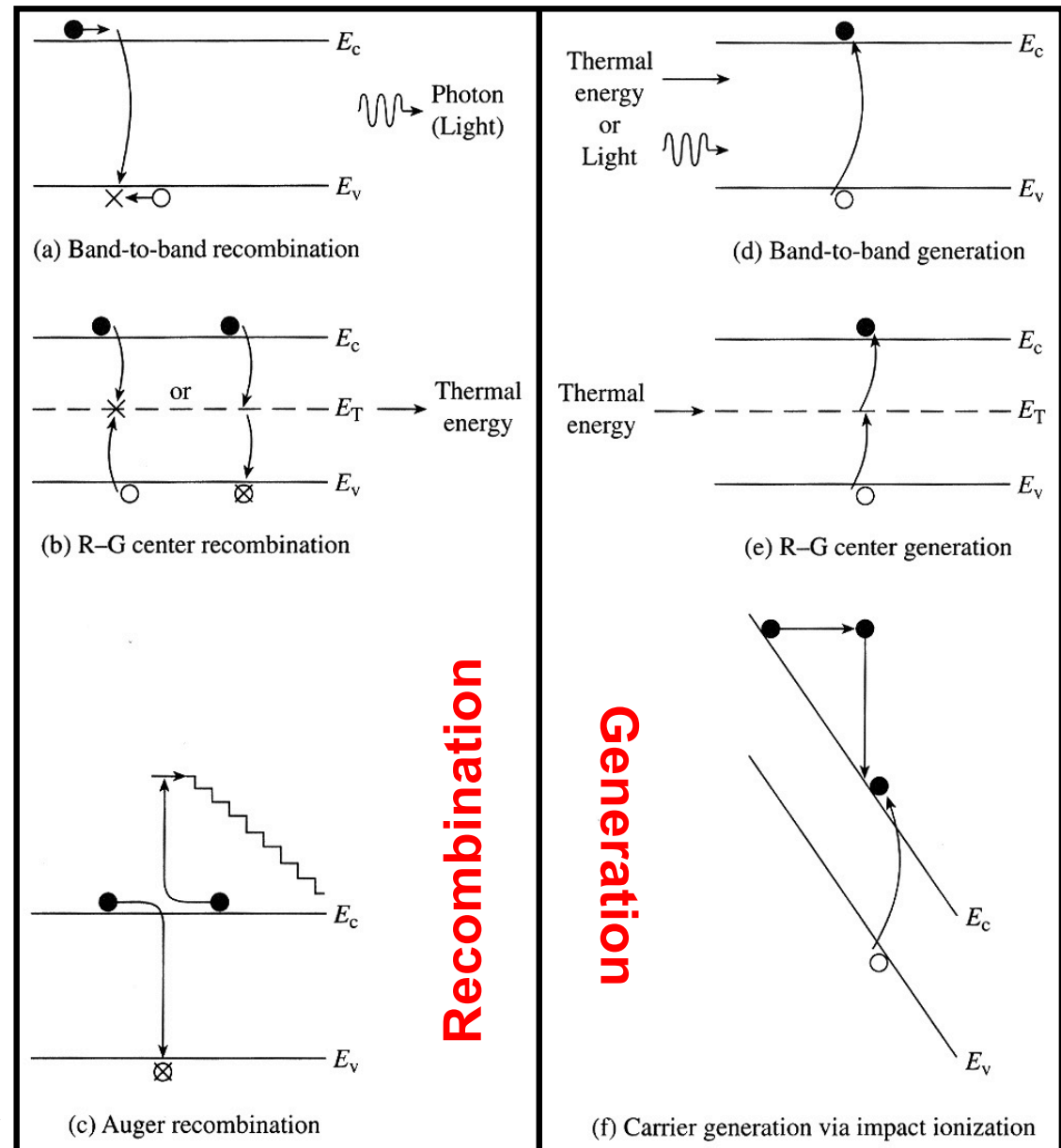
2.7.3 Continuity Equations

Literature: Pierret, Chapter 3.3-3.5.1, page 105-132

2.7.1 Generation/Recombination Processes

- **Band-to-band R-G** processes “only” involve electron in conduction band and hole in valence band
- **R-G center generation/recombination** requires R-G center (lattice defect or impurity which generates states in the bandgap)
- **Auger recombination** requires 3 carriers (either 2 holes and 1 electron or 2 electrons and 1 hole)
- All processes occur at all times (even in equilibrium), with the process having the highest rate dominating

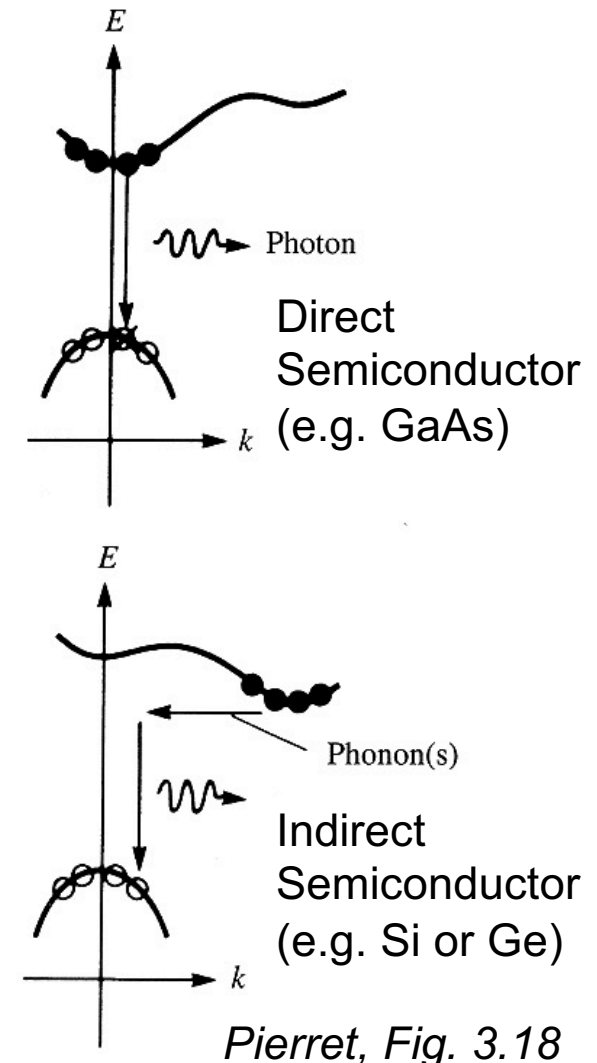
Pierret, Fig. 3.15



Momentum Considerations

Band-to-Band vs. R-G Center R-G Processes

- What is the dominant R-G process, i.e. the process proceeding at the fastest rate, in a particular system?
- When do band-to-band R-G processes dominate over R-G center R-G processes and vice versa?
- **Photons** carry energy but very little momentum
Phonons carry momentum but very little energy
- In **indirect semiconductors**, band-to-band recombination requires the emission/absorption of a phonon (for momentum conservation) in addition to photon emission, making this process negligible compared to R-G center recombination
- In **direct semiconductors**, band-to-band recombination proceeds at a much faster rate, making these materials suitable for light emitting devices, e.g. LEDs and lasers



Pierret, Fig. 3.18

Perturbation: Carrier Injection/Generation

$$pn > n_i^2$$

Processes:

- Photogeneration
- Operation of diode in forward direction
- Impact ionization

Perturbation: Carrier Extraction/Recombination

$$p n < n_i^2$$

Process:

- Operation of diode in reverse direction

Carrier Generation/Recombination

After switching off the distortion, the system returns to the **equilibrium state ($p n = n_i^2$)** with a characteristic time constant, i.e. excess carriers will recombine (in case $p n > n_i^2$) or additional carriers will be generated (in case $p n < n_i^2$)

Notation:	n_n	electron density in n-type semiconductor
	p_n	hole density in n-type semiconductor
	n_{n0}, p_{n0}	equilibrium carrier densities
	n_p	electron density in p-type semiconductor
	p_p	hole density in p-type semiconductor
	n_{p0}, p_{p0}	equilibrium carrier densities

Low-Level/High-Level Injection

Example: Injection by photogeneration ($\Delta n = \Delta p$) in **n-type** material

Low-Level Injection:

$$\Delta n = \Delta p \ll N_D$$

$$n_n = n_{n0} + \Delta n \approx n_{n0}$$

$$p_n = p_{n0} + \Delta p \approx \Delta p < n_{n0}$$

High-Level Injection:

$$\Delta n = \Delta p \geq N_D$$

$$n_n = n_{n0} + \Delta n > n_{n0}$$

$$p_n = p_{n0} + \Delta p \approx \Delta p > n_{n0}$$

Example: n-type Si, $N_D = n_{n0} = 10^{15} \text{ cm}^{-3}$, $p = n_i^2/n = 10^5 \text{ cm}^{-3}$

$$\Delta n = \Delta p = 10^{12} \text{ cm}^{-3}$$

$$n_n \approx 10^{15} \text{ cm}^{-3}$$

$$p_n \approx 10^{12} \text{ cm}^{-3}$$

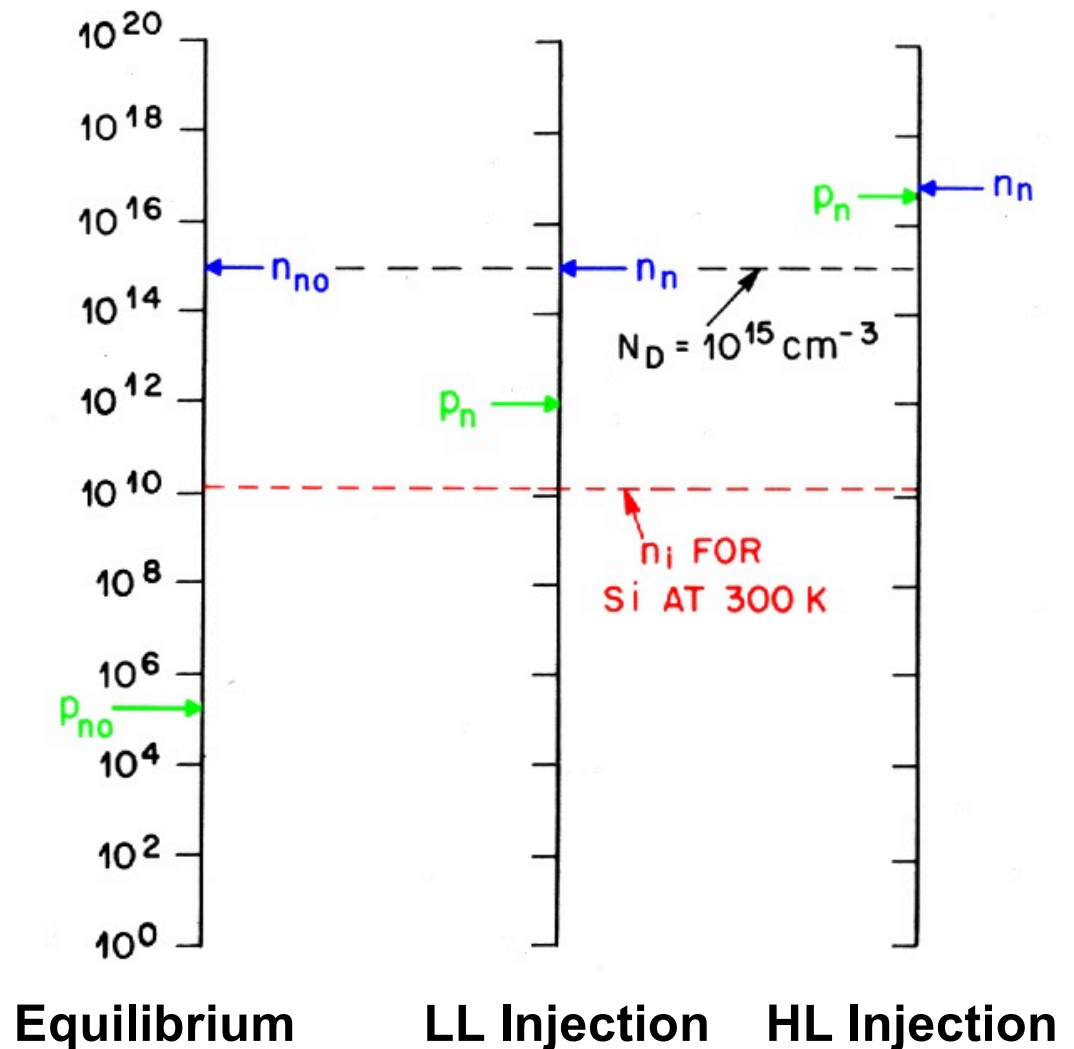
$$\Delta n = \Delta p = 10^{17} \text{ cm}^{-3}$$

$$n_n \approx 10^{17} \text{ cm}^{-3}$$

$$p_n \approx 10^{17} \text{ cm}^{-3}$$

Low-Level/High-Level Injection

- Example:
N-type semiconductor
 $N_D = 10^{15} \text{ cm}^{-3}$
- **Equilibrium:**
 $n_{n0} = N_D = 10^{15} \text{ cm}^{-3}$
 $p_{n0} = n_i^2/n = 10^5 \text{ cm}^{-3}$
- **Low-Level Injection:**
 $\Delta n = \Delta p = 10^{12} \text{ cm}^{-3}$
 $n_n \approx n_{n0} = 10^{15} \text{ cm}^{-3}$
 $p_n \approx \Delta p = 10^{12} \text{ cm}^{-3} < n_{n0}$
- **High-Level Injection:**
 $\Delta n = \Delta p = 10^{17} \text{ cm}^{-3}$
 $n_n \approx 10^{17} \text{ cm}^{-3} > n_{n0}$
 $p_n \approx \Delta p = 10^{17} \text{ cm}^{-3} > n_{n0}$



2.7.2 R-G Statistics

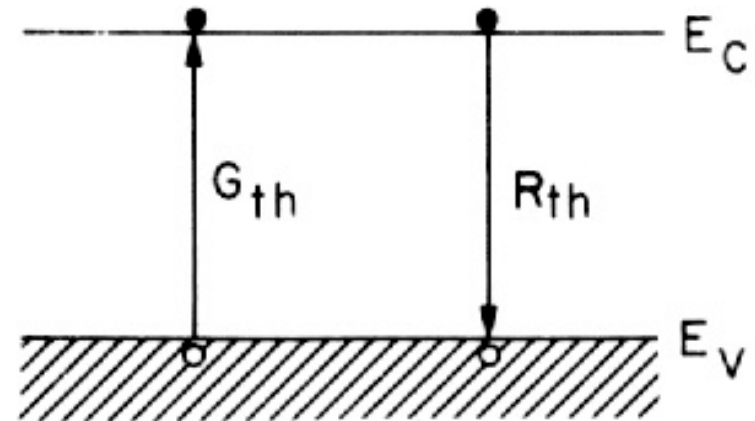
Band-to-Band Recombination

- Likely in **direct** (band gap) semiconductors, such as GaAs
- Unlikely in **indirect** (band gap) semiconductors, such as Si, because of required momentum conservation
- In **thermal equilibrium**, the thermal generation rate G_{th} equals the recombination rate R_{th} :

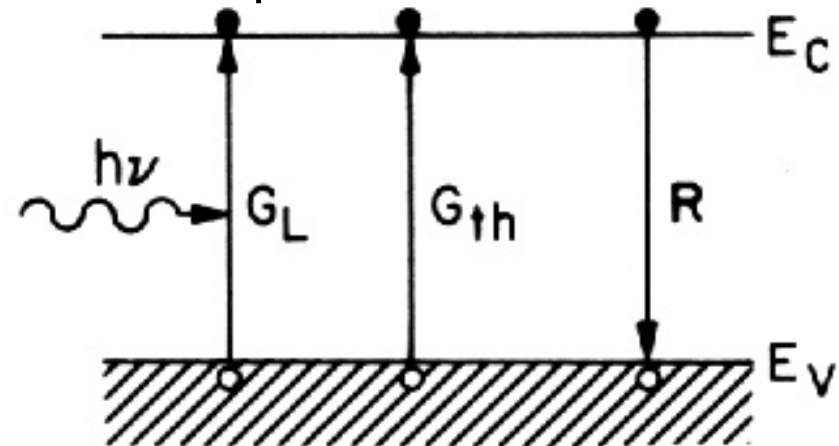
$$G_{th} = R_{th} = \beta n_{n0} p_{n0}$$

The thermal recombination rate is proportional to the electron and hole carrier densities with β being the proportionality factor

Equilibrium: *Sze, Fig. 3.10*



Non-Equilibrium:



Band-to-Band Recombination

- In **non-equilibrium**, the change in minority carrier concentration is:

$$\frac{dp_n(t)}{dt} = G_L + G_{th} - R$$

- Assuming steady state, $dp_n/dt = 0$, we obtain

$$G_L = R - G_{th} \equiv U$$

with the net-recombination rate U

- The recombination rate and the thermal generation rate are proportional to the available carrier densities (see equilibrium):

$$\begin{aligned} U &= \underbrace{\beta n_n p_n}_{=R} - \underbrace{\beta n_{n0} p_{n0}}_{=G_{th}} \\ &= \beta \left[(n_{n0} + \Delta n)(p_{n0} + \Delta p) - n_{n0} p_{n0} \right] \\ &= \beta [n_{n0} + p_{n0} + \Delta p] \Delta p \end{aligned}$$

- Assuming **low-level injection**, i.e. Δp and $p_{n0} \ll n_{n0}$, we obtain...

Band-to-Band Recombination

$$U = \beta n_{n0} \Delta p = \frac{p_n - p_{n0}}{\frac{1}{\beta n_{n0}}} = \frac{p_n - p_{n0}}{\tau_p}$$

- The **minority carrier lifetime** τ_p controls the recombination “velocity” after the distortion is switched off

$$\tau_p \equiv \frac{1}{\beta n_{n0}} \rightarrow \frac{dp_n(t)}{dt} = G_L - U = G_L - \frac{p_n - p_{n0}}{\tau_p} = G_L - \frac{\Delta p_n}{\tau_p}$$

- The minority carrier lifetime in case of the direct recombination is inversely proportional to the majority carrier equilibrium density n_{n0}
- Excess carriers recombine after switching off G_L ($G_L = 0$ in DE) with the time constant τ_p :

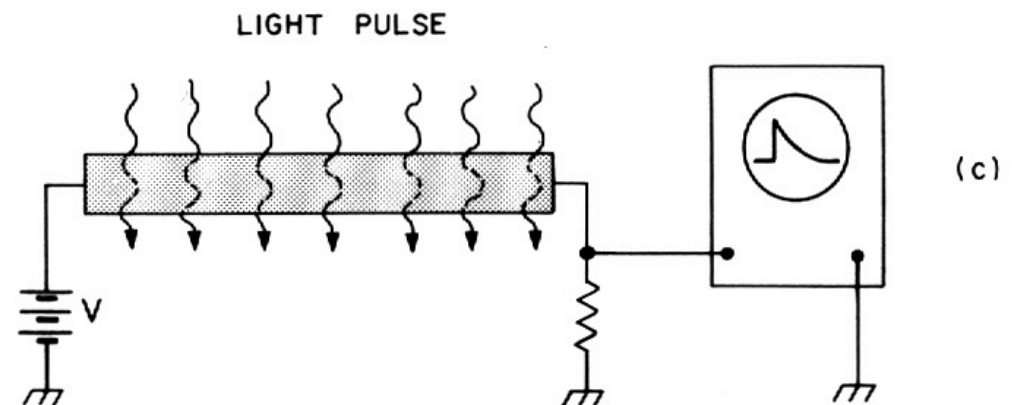
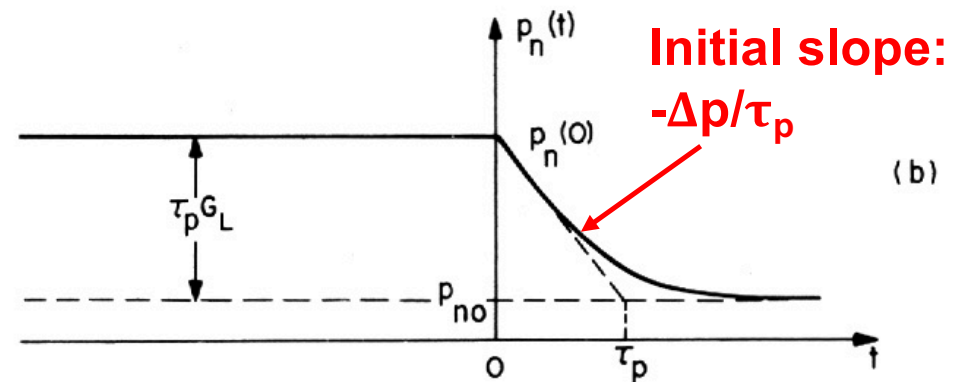
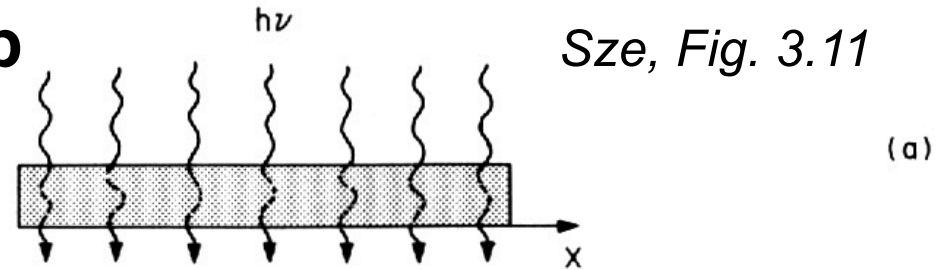
$$p_n(t) = p_{n0} + \tau_p G_L e^{-t/\tau_p}$$

- How can we measure the minority carrier lifetime?

Measurement of τ_p

- Illuminate semiconductor sample with light and measure its electrical resistance (or the voltage drop across the sample):
 - a. n-doped sample under constant illumination
 - b. Minority carrier concentration in steady state ($t < 0$) and after switching off the illumination
 - c. Experimental setup to determine the minority carrier lifetime τ_p

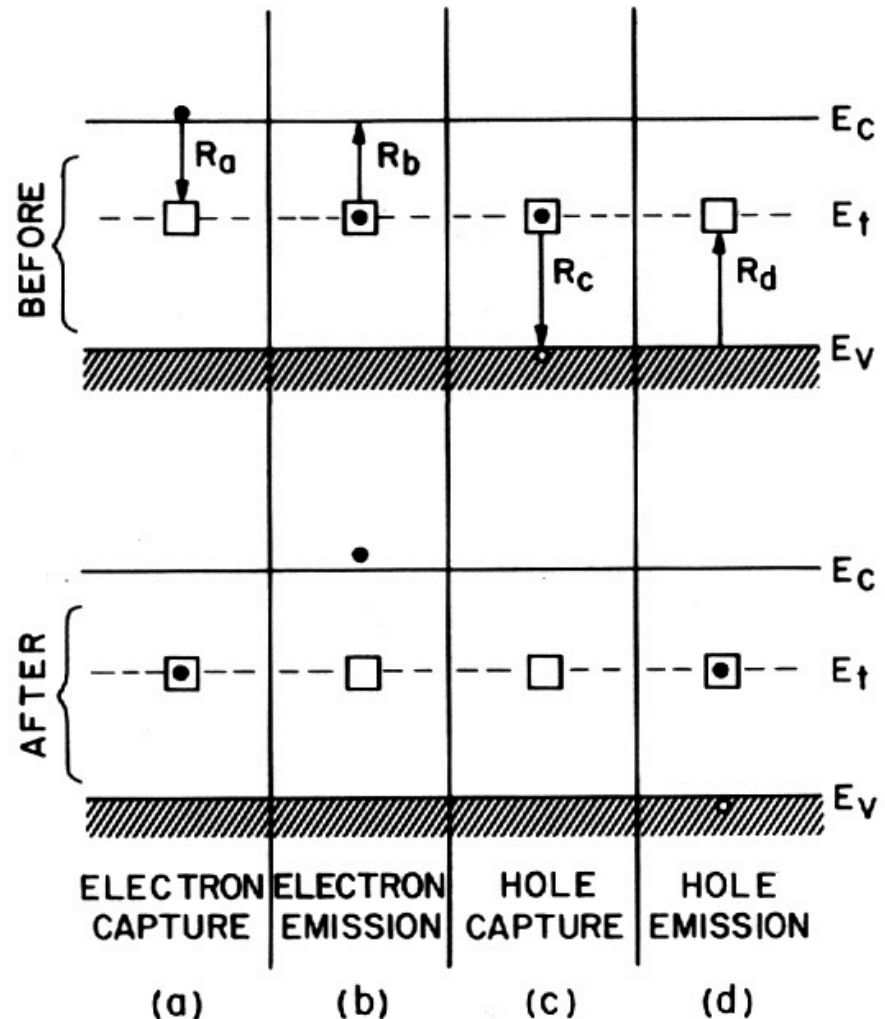
Sze, Fig. 3.11



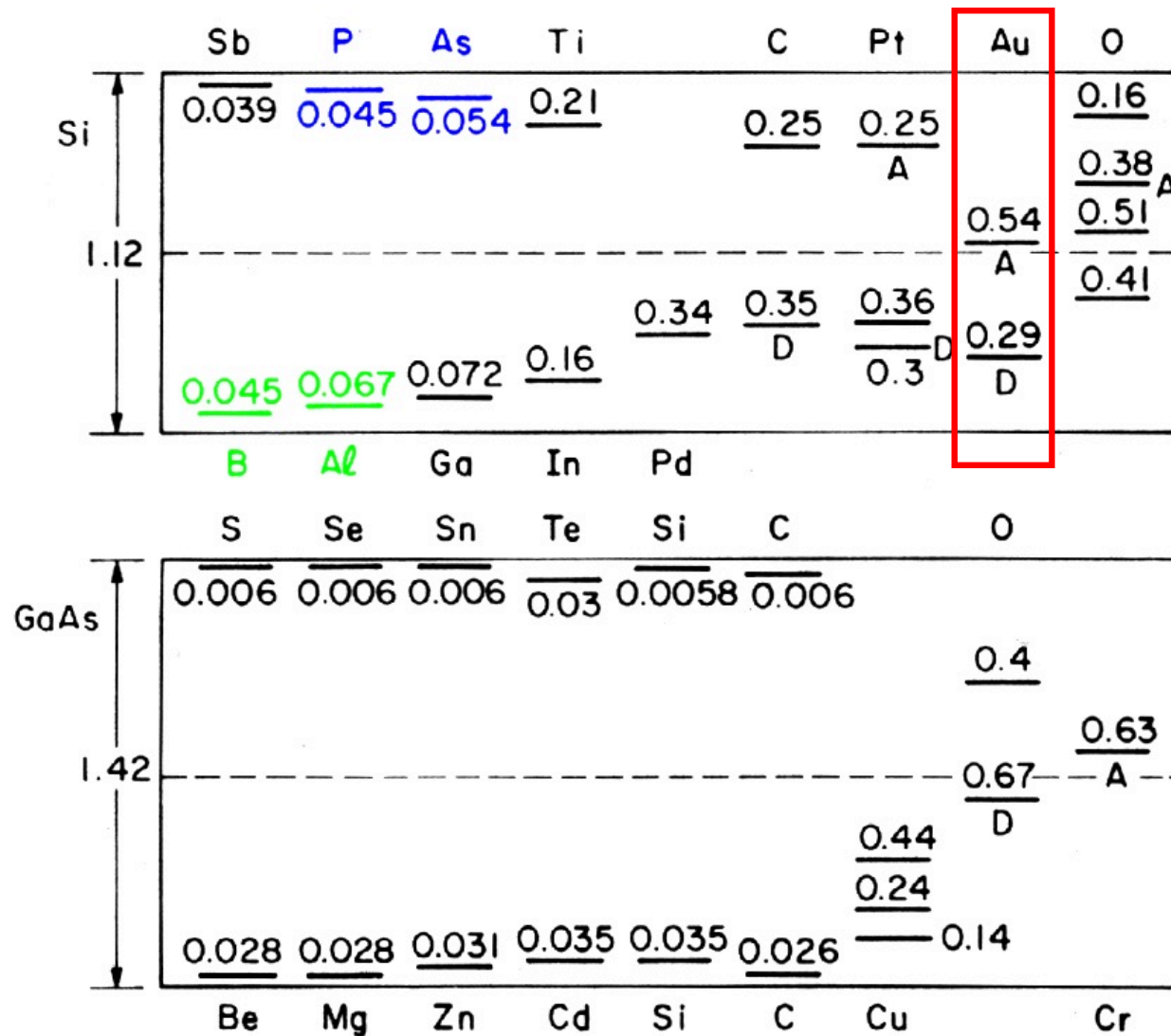
R-G Center Recombination

- Dominant recombination process in indirect semiconductors, such as Si
- Indirect recombination & generation of electron-hole pairs via localized energy states in the band gap, so called **recombination centers** or **R-G centers**
- The R-G centers are characterized by their **energy E_t** and their **density N_t**

Sze, Fig. 3.12



Location of “Impurity” Atom Energy States



Sze, Fig. 2.24

R-G Center Recombination

- In a more involved derivation, an expression similar to that of the band-to-band R/G case can be obtained for R-G center recombination/generation (again assuming minority carriers and low-level injection) in response to a perturbation:

$$\frac{dp_n(t)}{dt} = G_L - U = G_L - \frac{\Delta p_n}{\tau_p} \quad \text{and} \quad \frac{dn_p(t)}{dt} = G_L - \frac{\Delta n_p}{\tau_n}$$

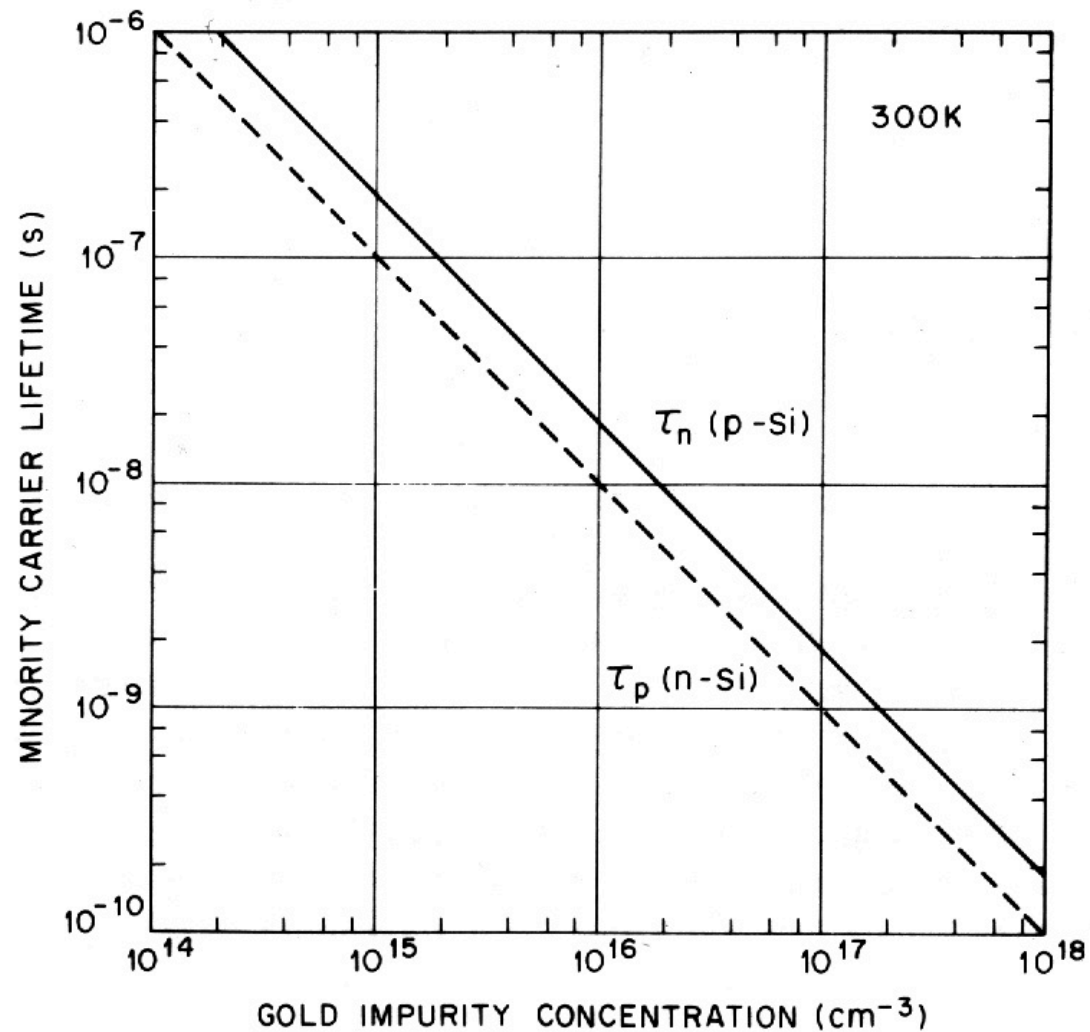
- The minority carrier lifetimes are now inversely proportional the R-G center concentration N_T :

$$\tau_n \propto \frac{1}{N_T} \quad \text{and} \quad \tau_p \propto \frac{1}{N_T}$$

- In case of arbitrary injection levels and for both carrier types in a non-degenerate semiconductor, we find:

$$U = \frac{(p_n n_n - n_i^2)}{\tau_p \left[p_n + n_i e^{(E_i - E_t)/kT} \right] + \tau_n \left[n_n + n_i e^{(E_t - E_i)/kT} \right]}$$

Example: Gold R-G Centers



2.7.3 Continuity Equations

“putting it all together”

Net change of carrier concentration due to currents and R/G:

$$\begin{aligned}
 \frac{\partial n}{\partial t} &= \underbrace{\frac{\partial n}{\partial t}\bigg|_{\text{drift}} + \frac{\partial n}{\partial t}\bigg|_{\text{diffusion}}}_{=\frac{1}{q} \nabla \cdot \vec{J}_n} + \underbrace{\frac{\partial n}{\partial t}\bigg|_{\text{thermal R-G}}}_{=-\frac{\Delta n}{\tau_n}} + \underbrace{\frac{\partial n}{\partial t}\bigg|_{\text{other processes}}}_{\text{e.g. } G_L} \\
 \frac{\partial p}{\partial t} &= \underbrace{\frac{\partial p}{\partial t}\bigg|_{\text{drift}} + \frac{\partial p}{\partial t}\bigg|_{\text{diffusion}}}_{=-\frac{1}{q} \nabla \cdot \vec{J}_p} + \underbrace{\frac{\partial p}{\partial t}\bigg|_{\text{thermal R-G}}}_{=-\frac{\Delta p}{\tau_p}} + \underbrace{\frac{\partial p}{\partial t}\bigg|_{\text{other processes}}}_{\text{e.g. } G_L}
 \end{aligned}$$

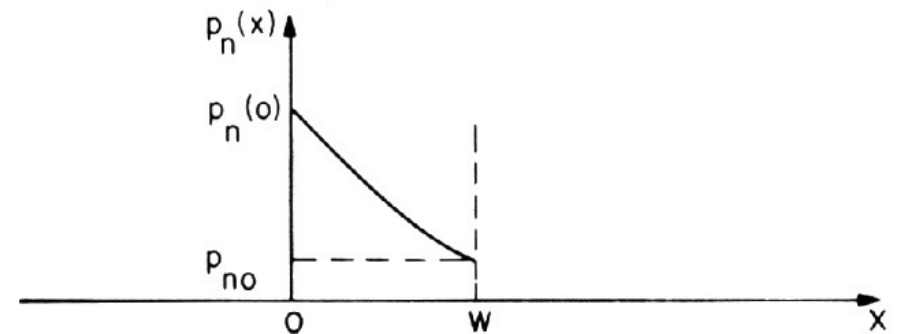
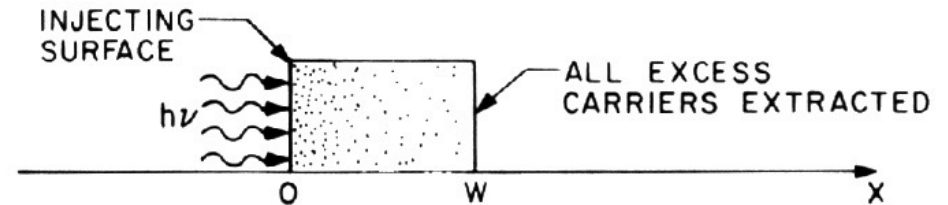
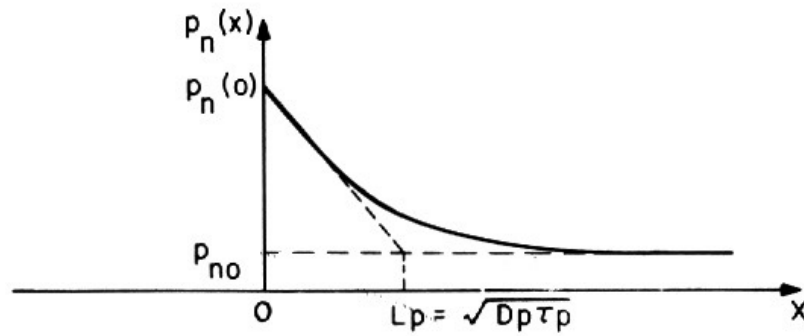
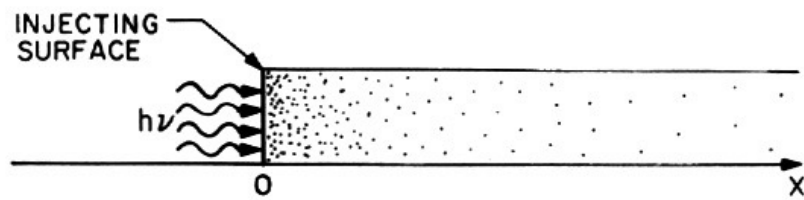
1-D Continuity Equations

For minority carriers in 1-D case: $J_p = q \mu_p p \mathcal{E} - q D_p \nabla p$
 $J_n = q \mu_n n \mathcal{E} + q D_n \nabla n$

$$\frac{\partial n_p(x,t)}{\partial t} = +n_p \mu_n \frac{\partial \mathcal{E}}{\partial x} + \mu_n \mathcal{E} \frac{\partial n_p}{\partial x} + D_n \frac{\partial^2 n_p}{\partial x^2} + G_L - \frac{n_p - n_{p0}}{\tau_n}$$

$$\frac{\partial p_n(x,t)}{\partial t} = \underbrace{-p_n \mu_p \frac{\partial \mathcal{E}}{\partial x} - \mu_p \mathcal{E} \frac{\partial p_n}{\partial x}}_{\text{Drift}} + \underbrace{D_p \frac{\partial^2 p_n}{\partial x^2}}_{\text{Diffusion}} + \underbrace{G_L}_{\text{Perturb.}} - \underbrace{\frac{p_n - p_{n0}}{\tau_p}}_{\text{thermal R-G}}$$

Continuity Equation – Sample Problems



Sze, Fig. 3.16

Calculate minority carrier distribution in 1-D semiconductor, without any electric field: **solve minority carrier diffusion equations!**

$$\frac{\partial n_p(x,t)}{\partial t} = D_n \frac{\partial^2 n_p}{\partial x^2} - \frac{n_p - n_{p0}}{\tau_n} + G_L$$

$$\frac{\partial p_n(x,t)}{\partial t} = D_p \frac{\partial^2 p_n}{\partial x^2} - \frac{p_n - p_{n0}}{\tau_p} + G_L$$